

ANSWER KEY

GATE EC — GATE EC 2026 Trend-Based Mock Paper
1

Q.No.	Type	Answer	Marks
1	2MCQ	B	2
2	2NAT	—	2
3	2MCQ	B	2
4	2MCQ	—	2
5	2NAT	—	2
6	2MCQ	B	2
7	3MCQ	—	3
8	2MCQ	—	2
9	3MCQ	—	3
10	2MCQ	—	2
11	2NAT	—	2
12	2NAT	A	2
13	2NAT	B	2
14	1NAT	B	1
15	1NAT	B	1
16	2MCQ	—	2
17	2MCQ	—	2
18	2MCQ	—	2
19	2MSQ	—	2
20	2MCQ	—	2
21	2MCQ	—	2
22	2NAT	—	2
23	2MCQ	—	2
24	1MCQ	B	1
25	1MCQ	B	1
26	1MCQ	B	1
27	1MCQ	B	1
28	1MCQ	B	1
29	1NAT	C	1
30	1MCQ	B	1
31	1MCQ	C	1
32	1MCQ	A	1
33	1NAT	A	1
34	1MCQ	B	1
35	1MCQ	A	1
36	1MCQ	A	1

37	1MCQ	B	1
38	1MCQ	A	1
39	1MCQ	A	1
40	1MCQ	B	1
41	1NAT	A	1
42	1MCQ	A	1
43	1NAT	A	1
44	1MCQ	A	1
45	1NAT	C	1
46	1MCQ	A	1
47	1NAT	D	1
48	1MCQ	B	1
49	1MCQ	B	1
50	2NAT	—	2
51	2NAT	—	2
52	2MCQ	—	2
53	2MCQ	—	2
54	2MCQ	—	2
55	1MCQ	B	1
56	2MSQ	—	2
57	1MCQ	A	1
58	2NAT	—	2
59	2NAT	—	2
60	2NAT	—	2
61	1MCQ	B	1
62	2MSQ	—	2
63	1MCQ	—	1
64	2MCQ	—	2
65	2MSQ	—	2

Q1. General Aptitude | 2 marks | 2MCQ
Sum of first 100 natural numbers:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q10

Q2. General Aptitude | 2 marks | 2NAT
The 12 musical notes are given as , #, , #, , #, , #, , #, and . Frequency of each note is 122 times the frequency of ...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q3

Q3. General Aptitude | 2 marks | 2MCQ
If train 100m crosses pole in 10s, speed:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q10

Q4. General Aptitude | 2 marks | 2MCQ
For a real number $x > 1$, $\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \frac{1}{x^4} = 1$ The value of x is

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 7 Q4

Q5. General Aptitude | 2 marks | 2NAT
Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify th...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q1

Q6. General Aptitude | 2 marks | 2MCQ
Choose synonym of 'trivial':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q10

Q7. General Aptitude | 3 marks | 3MCQ
The greatest prime factor of $(3199 - 3196)$ is

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 7 Q5

Q8. General Aptitude | 2 marks | 2MCQ
The following figures show three curves generated using an iterative algorithm. The total length of the curve generated ...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q4

Q9. General Aptitude | 3 marks | 3MCQ

Which one of the following plots represents $y = -||x|$, where x is a non-zero real number? Note: The figures shown are rep...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q5

Q10. General Aptitude | 2 marks | 2MCQ

If '++' denotes increasing order of intensity, then the meaning of the words [charm enamor bewitch] is analogous to [bored...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 7 Q1

Q11. General Aptitude | 2 marks | 2NAT

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above ...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q2

Q12. General Aptitude | 2 marks | 2NAT

Choose opposite of 'benevolent':

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q10

Q13. Engineering Mathematics | 2 marks | 2NAT

$i^4 = ?$

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q1

Q14. Engineering Mathematics | 1 marks | 1NAT

$P(A \cap B) = ?$

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q1

Q15. Engineering Mathematics | 1 marks | 1NAT

Integral of $1/x$ dx:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q1

Q16. Network, Signals & Systems | 2 marks | 2MCQ

A full adder and an XOR gate are used to design a digital circuit with inputs X, Y, and Z, and output F, as shown below....

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q24

Q17. Network, Signals & Systems | 2 marks | 2MCQ

Consider the unity-negative-feedback system shown in Figure (i) below, where gain $K = 0$. The root locus of this system is sh...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q26

Q18. Network, Signals & Systems | 2 marks | 2MCQ

The free electron concentration profile $n(x)$ in a doped semiconductor at equilibrium is shown in the figure, where the $p...$

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q26

Q19. Network, Signals & Systems | 2 marks | 2MSQ

As shown in the circuit, the initial voltage across the capacitor is 10 V, with the switch being open. The switch is the...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q35

Q20. Network, Signals & Systems | 2 marks | 2MCQ

A 50 lossless transmission line is terminated with a load Z_L of $(50 - j75)$. If the average incident power on the line i...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q64

Q21. Network, Signals & Systems | 2 marks | 2MCQ

A stick of length one meter is broken at two locations at distances of 1 and 2 from the origin (0), as shown in the figu...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q7

Q22. Network, Signals & Systems | 2 marks | 2NAT

The diode in the circuit shown below is ideal. The input voltage (in Volts) is given by $= 10 \sin 100t$, where time is in s...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q61

Q23. Network, Signals & Systems | 2 marks | 2MCQ

The intrinsic carrier concentration of a semiconductor is $2.5 \times 10^{16} /m^3$ at 300 K. If the electron and hole mobilities a...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q46

Q24. Electronic Devices | 1 marks | 1MCQ

SCR has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q9

Q25. Electronic Devices | 1 marks | 1MCQ

n-type majority carriers:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q3

Q26. Electronic Devices | 1 marks | 1MCQ

MOSFET is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q8

Q27. Electronic Devices | 1 marks | 1MCQ
Zener diode used for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q9

Q28. Analog Circuits | 1 marks | 1MCQ
Common emitter has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q4

Q29. Analog Circuits | 1 marks | 1NAT
Ideal op-amp open-loop gain:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 EC Session 1 Q2

Q30. Analog Circuits | 1 marks | 1MCQ
FET is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q3

Q31. Analog Circuits | 1 marks | 1MCQ
For a common emitter amplifier, the voltage gain is approximately:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 EC Session 1 Q3

Q32. Analog Circuits | 1 marks | 1MCQ
RC oscillator frequency:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q9

Q33. Digital Circuits | 1 marks | 1NAT
ROM is:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q4

Q34. Digital Circuits | 1 marks | 1MCQ
D flip-flop follows input after:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q5

Q35. Digital Circuits | 1 marks | 1MCQ
4-bit binary counter counts:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q4

Q36. Digital Circuits | 1 marks | 1MCQ
TTL stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q3

Q37. Control Systems | 1 marks | 1MCQ
Transfer function is defined:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q5

Q38. Control Systems | 1 marks | 1MCQ
Nyquist criterion uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q6

Q39. Control Systems | 1 marks | 1MCQ
Routh-Hurwitz determines:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q7

Q40. Control Systems | 1 marks | 1MCQ
Lead compensator:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q6

Q41. Communication Systems | 1 marks | 1NAT
ASK modulation varies:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q5

Q42. Communication Systems | 1 marks | 1MCQ
FM has:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q5

Q43. Communication Systems | 1 marks | 1NAT
PCM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q4

Q44. Communication Systems | 1 marks | 1MCQ
AM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q6

Q45. Electromagnetics | 1 marks | 1NAT
Maxwell has how many equations:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 EC Session 1 Q8

Q46. Electromagnetics | 1 marks | 1MCQ
Characteristic impedance Z_0 =:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q7

Q47. Electromagnetics | 1 marks | 1NAT
Skin depth increases with:

Answer: D

Correct Answer: D

Source: Adapted from GATE 2023 EC Session 1 Q6

Q48. Electromagnetics | 1 marks | 1MCQ
Waveguide supports:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q7

Q49. Control Systems | 1 marks | 1MCQ
Transfer function is defined:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q5

Q50. Network, Signals & Systems | 2 marks | 2NAT
 $s(s+1)(s+2)$ $R(s)$ + $E(s)$ $G(s)$ $Y(s)$ - $R(s)$, $Y(s)$, and $E(s)$ are the Laplace transforms of $r(t)$, $y(t)$, and $e(t)$, resp...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q13

Q51. Network, Signals & Systems | 2 marks | 2NAT

A 50 lossless transmission line is terminated with a load Z_L of $(50 - j75)$. If the average incident power on the line is...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q64

Q52. Network, Signals & Systems | 2 marks | 2MCQ

In the circuit shown below, the transistors M_1 and M_2 are biased in saturation. Their small signal transconductances are...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q51

Q53. Network, Signals & Systems | 2 marks | 2MCQ

Let (x, y, z, t) and $u(x, y, z, t)$ represent density and velocity, respectively, at a point (x, y, z) and time t . Assume is...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q25

Q54. Network, Signals & Systems | 2 marks | 2MCQ

Consider a continuous-time finite-energy signal $x(t)$ whose Fourier transform vanishes outside the frequency interval $[-f_m, f_m]$...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q38

Q55. Analog Circuits | 1 marks | 1MCQ

Common emitter has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q4

Q56. Network, Signals & Systems | 2 marks | 2MSQ

Consider a system represented by the block diagram shown below. Which of the following signal flow graphs represent(s) the system?

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q56

Q57. Communication Systems | 1 marks | 1MCQ

PCM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q4

Q58. Network, Signals & Systems | 2 marks | 2NAT

A full scale sinusoidal signal is applied to a 10-bit ADC. The fundamental signal component in the ADC output has a normalized magnitude of...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q46

Q59. Network, Signals & Systems | 2 marks | 2NAT

Consider the unity-negative-feedback system shown in Figure (i) below, where $G(s) = \frac{1}{s(s+1)}$. The root locus of this system is shown in Figure (ii) below. The value of K for which the system is marginally stable is...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q26

Q60. Network, Signals & Systems | 2 marks | 2NAT
A satellite attitude control system, as shown below, has a plant with transfer function $G(s) = 1$ cascaded with a compensator...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q38

Q61. Engineering Mathematics | 1 marks | 1MCQ
 $P(A \cap B) = ?$

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q1

Q62. Network, Signals & Systems | 2 marks | 2MSQ
Which of the following can be used as an n-type dopant for silicon? Select the correct option(s).

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q31

Q63. Network, Signals & Systems | 1 marks | 1MCQ
Sequence the following sentences (P, Q, R, S) in a coherent passage: P: Shifu's student exclaimed, "Why do you run since..."

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 7 Q6

Q64. Network, Signals & Systems | 2 marks | 2MCQ
The relationship between any N-length sequence $x[n]$ and its corresponding N-point discrete Fourier transform $X[k]$ is defined...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q59

Q65. Network, Signals & Systems | 2 marks | 2MSQ
Which of the following statements is/are TRUE with respect to ideal MOSFET-based DC-coupled single-stage amplifiers having...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q30